

Supplementary Information

Whose Regulation Does the Model Know? Reliability of LLM-Retrieved Regulatory Information for Public Environmental Management in 25 Countries

Reproducibility. Every data-driven table below (Tables [S1–S7](#)) is regenerated from the raw confirmatory artifacts by `code/analysis/make_supplement_tables.py`; no value is hand-entered. Scoring is adjudicated by code against the official register wherever the correct answer is a checkable value (T1, T2, T3), and by the mean of a three-vendor judge panel (Gemini 2.5 Pro, Claude Sonnet 4.6, DeepSeek-V3) elsewhere; T5 retains the scores of the collection judge (`gpt-5-mini-2025-08-07` for the pre-specified fifteen countries, `claude-haiku-4-5-20251001` for the ten added in the extension). The analysed confirmatory corpus comprises 8,244 scored cells (6,573 English-prompt; 1,671 native-language) over 25 countries and 14 deployment stacks, after removing the 56 Egyptian T1 cells for which the registry holds no key. All inputs, the scoring code, the ground-truth registry, and the anonymised responses will be released on publication under CC-BY-4.0 (data/prompts) and MIT (code).

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S1 Pre-specification, design, and country selection

The hypotheses, primary models, persona protocol, exclusion criteria, multiple-comparison corrections, and the country sampling frame were specified in advance of confirmatory data collection. The pre-specified design covers 15 countries; the sample was subsequently extended, *post registration*, to 25 countries to gain power for the development gradient (H1) and the corpus-mechanism tests (H4) and to multiply the within-country language pairs (H2). Throughout the manuscript and this supplement, every effect computed on the 25-country sample is reported alongside its pre-specified 15-country value, and no pre-specified conclusion is revised on the basis of the extension alone (Section S8).

Countries were selected by theoretical sampling triangulated across three independent classifications: the Global North/South distinction, taken from UNCTAD’s developed/developing economies classification (operationalising the coloniality-of-knowledge framing), the Joshi et al. (2020) six-class linguistic-resource taxonomy (Joshi et al., 2020), and World Bank income groups. The pre-specified 15 (Set “P” in Table S4) span four world regions, five Joshi classes, and four income groups; the 10 extension countries (Set “E”) comprise six additional Global North references (United Kingdom, Canada, Australia, South Korea, France, Italy) chosen to thicken the high-development end of the gradient, and four additional native-language-pair countries (Colombia, Chile, Angola, Portugal), of which Portugal is Global North; the extension therefore adds seven Global North and three Global South countries. UNCTAD states that developed economies “broadly comprise Northern America and Europe, Israel, Japan, the Republic of Korea, Australia, and New Zealand”, developing economies being the remainder; all 25 countries fall on the side that rule assigns them, South Korea included. The full design rationale is in docs/supplement/s1_design_rationale.md.

S2 Deployment-stack roster and serving configuration

Unit of analysis. We evaluate each model *as served* through a fixed access stack — the tuple of model weights, vendor, serving infrastructure, and API/endpoint layer — rather than idealized weights in isolation, because a public manager interacts with the served stack and any measured gap is a property of that stack. Table S1 lists the 14 stacks with their exact API/tag strings and execution venues.

Table S1. Deployment-stack roster. Each model is evaluated *as served* through a fixed access stack; the API/tag string and execution venue are part of what each label denotes. Parameter counts are vendor-reported (dense or total for MoE; closed-frontier counts undisclosed). *N* and mean are realized English-prompt confirmatory coverage.

Model	Vendor	Tier	API / tag string	Venue	<i>N</i>	Mean
DeepSeek-V3	DeepSeek	A	deepseek-chat	DeepSeek API	496	0.687
Command R+	Cohere	A	command-r-plus-08-2024	Cohere (trial)	496	0.530

Model		Vendor	Tier	API / tag string	Venue	<i>N</i>	Mean
Llama 3.3 70B		Meta	A	llama-3.3-70b-versatile	Groq (free)	391	0.511
Llama 4 Scout		Meta	A	meta-llama/ llama-4-scout-17b-16e-instruct	Groq (free)	496	0.498
GPT-OSS 120B		OpenAI	A	openai/gpt-oss-120b	Groq (free)	257	0.484
Qwen3 32B		Alibaba	B	qwen/qwen3-32b	Groq (free)	494	0.562
Qwen3 14B		Alibaba	B	qwen3:14b	Ollama (local)	496	0.485
Phi-4 14B		Microsoft	B	phi4	Ollama (local)	496	0.477
Llama 3.1 8B		Meta	C	llama-3.1-8b-instant	Groq (free)	496	0.438
Cabra-Mistral 7B		botbot-ai	C	cabra-mistral-7b	Ollama (local)	492	0.351
Gemini Flash	2.5	Google	D	gemini-2.5-flash	Google AI (free)	486	0.634
GPT-5-mini		OpenAI	D	gpt-5-mini	OpenAI API	495	0.626
Claude 4.5	Haiku	Anthropic	D	claude-haiku-4-5	Anthropic API	486	0.591
GPT-5		OpenAI	E	gpt-5	OpenAI API	496	0.674

Determinism and seed handling. All stacks were queried at temperature 0.3 with the maximum-determinism configuration each provider documents, not an identical parameter surface, because the surfaces differ. Seed support is heterogeneous: locally served stacks (Ollama) honour a fixed seed; OpenAI exposes an advisory seed; Anthropic’s Messages API exposes no seed parameter; and for the remaining hosted endpoints a seed is either advisory or not exposed. For every call we record the requested seed and a `seed_status` field (`sent`, `logged-only`, or `not-supported`) in the released run metadata, so that the determinism actually achieved is auditable rather than assumed. Two stochastic replications per (prompt, model, persona) cell estimate within-cell variability.

Acquisition window. Served-model behaviour can drift as vendors update infrastructure, so the reported accuracies are point-in-time estimates relative to the collection window. The pinned tag, version string, and acquisition timestamp are recorded for every call. Confirmatory collection ran from 2026-04-26 to 2026-06-12 (UTC), spanning the pilot, the pre-specified 15-country collection, and the post-hoc extension to 25 countries. Table S2 gives the first and last non-error call timestamp per provider together with the official API documentation consulted; the full per-call timestamps are released with the dataset.

Table S2. Per-provider acquisition window and official API documentation. Reported accuracies are point-in-time estimates relative to this window; first/last are the earliest and latest non-error call timestamps (UTC) recorded in the run logs. The window spans the pilot, the pre-specified 15-country collection, and the extension.

Provider (venue)	First (UTC)	Last (UTC)	API documentation
Anthropic API	2026-04-27	2026-06-12	https://docs.anthropic.com/en/api/messages
Cohere (trial)	2026-04-26	2026-06-11	https://docs.cohere.com/reference/chat
DeepSeek API	2026-04-26	2026-06-11	https://api-docs.deepseek.com
Google AI (free)	2026-04-27	2026-06-11	https://ai.google.dev/api
Groq (free)	2026-04-26	2026-06-11	https://console.groq.com/docs/api-reference
Ollama (local)	2026-04-27	2026-06-12	https://github.com/ollama/ollama/blob/main/docs/api.md
OpenAI API	2026-04-27	2026-06-12	https://platform.openai.com/docs/api-reference/chat
All providers	2026-04-26	2026-06-12	—

S3 Coverage

Per-cell coverage is not uniform across the roster: free and rate-limited endpoints returned fewer admissible responses than the metered APIs, and one candidate (Gemini 2.5 Flash on early native batches) and a sixth judge (Command R+) hit provider quotas. We report realized coverage rather than silently dropping under-covered cells. Table S3 gives the model $\times$ task English-prompt matrix (column totals equal the by-task counts in the main text); Table S4 gives per-country English, native, and total counts with tier and pre-specification set; Table S5 lists the native-language pairs used for H2. The complete model $\times$ country $\times$ task coverage CSV, with the quota or exclusion reason for every shortfall, is included in the release.

Table S3. Coverage matrix: realized English-prompt confirmatory responses per model $\times$ task (T1 standard, T2 local datum, T3 health synthesis, T4 instruments, T5 recommendation). No cell is silently dropped; shortfalls are quota-driven (Methods).

Model	T1	T2	T3	T4	T5	Total
GPT-5	96	100	100	100	100	496
DeepSeek-V3	96	100	100	100	100	496
Command R+	96	100	100	100	100	496

Model	T1	T2	T3	T4	T5	Total
Phi-4 14B	96	100	100	100	100	496
Qwen3 14B	96	100	100	100	100	496
Llama 4 Scout	96	100	100	100	100	496
Llama 3.1 8B	96	100	100	100	100	496
GPT-5-mini	96	100	100	99	100	495
Qwen3 32B	96	100	100	99	99	494
Cabra-Mistral 7B	95	97	100	100	100	492
Gemini 2.5 Flash	94	98	98	98	98	486
Claude Haiku 4.5	94	98	98	98	98	486
Llama 3.3 70B	78	81	80	76	76	391
GPT-OSS 120B	54	61	53	40	49	257
Total	1279	1335	1329	1310	1320	6573

Table S4. Coverage by country: tier (GN/GS), sample set (P = pre-specified 15; E = extension), and realized confirmatory responses (English, native-language, total).

Country	Tier	Set	English	Native	Total
Colombia	GS	E	260	251	511
Chile	GS	E	260	241	501
Portugal	GN	E	261	240	501
Angola	GS	E	260	240	500
Argentina	GS	P	280	139	419
India	GS	P	279	138	417
Mexico	GS	P	269	140	409
Brazil	GS	P	260	141	401
Peru	GS	P	256	141	397
Germany	GN	P	279	0	279
Indonesia	GS	P	279	0	279
Japan	GN	P	279	0	279
Kenya	GS	P	279	0	279
Bangladesh	GS	P	278	0	278
United Kingdom	GN	E	265	0	265
Canada	GN	E	260	0	260
Australia	GN	E	260	0	260
France	GN	E	260	0	260
Italy	GN	E	260	0	260
South Korea	GN	E	258	0	258
Nigeria	GS	P	257	0	257

Country	Tier	Set	English	Native	Total
Philippines	GS	P	256	0	256
United States	GN	P	252	0	252
South Africa	GS	P	242	0	242
Egypt	GS	P	224	0	224
Total (25)			6573	1671	8244

Table S5. Native-language confirmatory responses by language and country (H2 within-country English/native contrast).

Native language (Joshi)	Country	<i>N</i> (native)
Portuguese (4)	Portugal	240
	Angola	240
	Brazil	141
Spanish (5)	Colombia	251
	Chile	241
	Peru	141
	Mexico	140
	Argentina	139
Hindi (4)	India	138

S4 Country covariates

Table S6 reports the developmental and corpus-representation covariates used in the mechanism analysis (H4). Language-corpus measures (Joshi class; mC4/OSCAR sizes reported in the main text) index how much text exists *in* a country’s language and are the candidate channel for the native-language penalty (H2). Country-coverage measures (English-Wikipedia article byte length, Wikidata sitelinks, Wikidata statements) index how broadly the encyclopedic corpus is *about* a country, independent of language, and are the candidate channel for the geographic gap (H1/H4). Because the geographic gap is measured *in English*, a residual North/South gap cannot be a language-corpus effect, which is why H4 isolates country coverage for H1 and language-corpus size for H2. All Wikidata entities are identified by a resolvable QID and were retrieved within the collection window.

Table S6. Country covariates. HDI: UNDP HDR 2023–24 (2022 data). Joshi: linguistic-resource class of the dominant official language (Joshi et al., 2020). Corpus measures from Wikimedia/Wikidata (entity QID resolvable): English-article byte length, Wikidata sitelinks (language editions), and Wikidata statements.

Country	Tier	HDI	Joshi	EN-wiki bytes	Sitelinks	WD stmts
Germany	GN	0.950	5	207,810	403	1402

Country	Tier	HDI	Joshi	EN-wiki bytes	Sitelinks	WD stmts
Australia	GN	0.946	5	233,095	395	1674
United Kingdom	GN	0.940	5	368,888	391	1085
Canada	GN	0.935	5	281,016	393	1346
South Korea	GN	0.929	4	275,291	350	746
United States	GN	0.927	5	383,914	424	1710
Japan	GN	0.920	5	205,844	414	995
France	GN	0.910	5	279,498	413	1196
Italy	GN	0.906	4	305,234	405	1039
Portugal	GN	0.874	4	324,038	369	751
Chile	GS	0.860	5	232,108	379	689
Argentina	GS	0.849	5	267,662	369	881
Mexico	GS	0.781	5	263,757	398	1286
Peru	GS	0.762	5	239,544	332	867
Brazil	GS	0.760	4	301,844	384	1151
Colombia	GS	0.758	5	303,401	363	774
Egypt	GS	0.728	5	284,547	387	757
South Africa	GS	0.717	5	270,861	365	750
Indonesia	GS	0.713	3	226,349	362	1005
Philippines	GS	0.710	5	493,981	331	963
Bangladesh	GS	0.670	3	275,786	343	814
India	GS	0.644	5	310,681	399	1679
Kenya	GS	0.601	5	216,189	334	699
Angola	GS	0.591	4	180,098	322	676
Nigeria	GS	0.548	5	266,041	349	1194

S5 Ground-truth registry (T1)

The validity of the geographic gradient (H1) requires that ground truth be equally available and equally verifiable across countries. For T1 (the binding annual PM_{2.5} standard) we anchored every country to an official primary source by documentary audit: the source text was retrieved, the value was read verbatim, and the entry records the value, status, source-of-record, a resolvable URL, and the retrieval method (Table S7). Three countries have *no* national PM_{2.5} standard, confirmed from the official text (Nigeria’s NESREA regulates PM₁₀ only; Argentina has no binding federal value; Angola has no national air-quality legislation); in those cases a faithful model must decline to state a non-existent threshold rather than fabricate one. This provenance is more reproducible than expert assertion: any reader can re-verify each value from the cited official source. One entry has no value: Egypt, whose official Arabic text could not be retrieved, so nothing is stored and Egypt is excluded from T1. Angola’s no-standard status rests on a World Bank/WHO country note rather than on a national instrument, because no national air-quality instrument exists to cite. Every other value

was confirmed against the primary text. The United Kingdom entry was corrected on 2026-09-21 from 10 (the English 2040 target) to 20 $\mu\text{g}/\text{m}^3$, the limit value in force since 2020, after checking DEFRA’s national objectives table; and the Bangladesh entry was corrected the same day from 15 (S.R.O. 220-Law/2005) to 35 $\mu\text{g}/\text{m}^3$ (Air Pollution (Control) Rules 2022, S.R.O. 255-Law/2022), after an external reader flagged the revision. Because the registry moved under scrutiny, every number in the manuscript is tied to a registry version: the released files carry SHA-256 prefixes `t1_registry` b0e5d9e132aab684, `t2_registry` 703dcd8122f88448, `t3_registry` 902de1bc6af893de, `t4_reference_set` 77bca4679aedc93a, and the analysis pipeline prints them.

Table S7. Ground-truth registry for T1 (binding annual $\text{PM}_{2.5}$ standard), all 25 countries, anchored to official primary sources by documentary audit. All entries are of source type *official primary* unless noted. Status *verified* = value read from the official text; *no standard* = official confirmation that no national standard exists; *partial* = the value could not be confirmed in an accessible official source (Bangladesh: a 2022 revision is known; Egypt: no value extracted, excluded from T1). Full URLs and verbatim excerpts are in the released registry JSONL.

Country	Value	Status	Official source
Angola	NO national air-quality legislation / standard	no standard	No national ambient air quality regulation (World Bank/WHO note)
Argentina	NO single binding federal $\text{PM}_{2.5}$ standard (Ley 20.284/1973 framework; $\text{PM}_{2.5}$ limits set subnationally)	no standard	Ley 20.284/1973
Australia	8 $\mu\text{g}/\text{m}^3$ (annual; 7 target 2025)	verified	National Environment Protection (Ambient Air Quality) Measure
Bangladesh	35 $\mu\text{g}/\text{m}^3$ (annual; 24-h 65) per Air Pollution (Control) Rules 2022, S.R.O. 255-Law/2022; superseded 15 (S.R.O. 220-Law/2005)	verified	Air Pollution (Control) Rules 2022 (S.R.O. 255-Law/2022), Schedule of ambient standards; confirmed in DoE, Ambient Air Quality in Bangladesh (2024), Table 3
Brazil	17 $\mu\text{g}/\text{m}^3$ (annual, PI-2)	verified	CONAMA 506/2024 Annex I, DOU 09/07/2024 Ed.130 p.133
Canada	8.8 $\mu\text{g}/\text{m}^3$ (annual, CAAQS 2020)	verified	Canadian Ambient Air Quality Standards (CCME)
Chile	20 $\mu\text{g}/\text{m}^3$ (annual)	verified	D.S. 12/2011 MMA Chile
Colombia	25 $\mu\text{g}/\text{m}^3$ (annual, in force since 2018; 15 from 2030; 24h=37)	verified	Resolucion 2254/2017 MinAmbiente Tabla 1 & 2 (verbatim official scan)
Egypt	$\text{PM}_{2.5}$ limit set under Law 4/1994 (amended 9/2009, 105/2015) Executive Regulations (official text in Arabic; value not extracted from an accessible official source)	partial	Environmental Law 4/1994 (EEAA)

Country	Value	Status	Official source
France	25 $\mu\text{g}/\text{m}^3$ (annual, in force; 10 from 2030)	verified	EU Ambient Air Quality Directive 2008/50/EC; (EU) 2024/2881
Germany	25 $\mu\text{g}/\text{m}^3$ (annual, in force 2026; 10 from 2030)	verified	EU Ambient Air Quality Directive 2008/50/EC; revised (EU) 2024/2881
India	40 $\mu\text{g}/\text{m}^3$ (annual)	verified	CPCB National Ambient Air Quality Standards (2009)
Indonesia	15 $\mu\text{g}/\text{m}^3$ (annual)	verified	PP No. 22/2021 Lampiran VII (Indonesia)
Italy	25 $\mu\text{g}/\text{m}^3$ (annual, in force; 10 from 2030)	verified	EU Ambient Air Quality Directive 2008/50/EC; (EU) 2024/2881
Japan	15 $\mu\text{g}/\text{m}^3$ (annual)	verified	Ministry of the Environment Japan, Environmental Quality Standards (2009)
Kenya	35 $\mu\text{g}/\text{m}^3$ (annual; 24h 75)	verified	First Schedule, EMCA (Air Quality) Regulations 2014 (Kenya Gazette Vol. CXVI No.46)
Mexico	10 $\mu\text{g}/\text{m}^3$ (annual)	verified	NOM-025-SSA1-2021, Diario Oficial de la Federacion 27/10/2021
Nigeria	NO national $\text{PM}_{2.5}$ standard (PM10 annual=60 $\mu\text{g}/\text{m}^3$, 24h=150)	no standard	NESREA National Environmental (Air Quality Control) Regulations, Schedule XIII (Ambient Air Quality Standards)
Peru	25 $\mu\text{g}/\text{m}^3$ (annual)	verified	D.S. 003-2017-MINAM (Peru)
Philippines	25 $\mu\text{g}/\text{m}^3$ (annual, 1-year guideline)	verified	DENR DAO (RA 8749 Clean Air Act); 25 $\mu\text{g}/\text{Nm}^3$ annual / 50 $\mu\text{g}/\text{Nm}^3$ 24h
Portugal	25 $\mu\text{g}/\text{m}^3$ (annual, in force; 10 from 2030)	verified	EU Ambient Air Quality Directive 2008/50/EC; (EU) 2024/2881
South Africa	20 $\mu\text{g}/\text{m}^3$ (annual)	verified	SA NAAQS under NEMAQA (Act 39/2004), 2012
South Korea	15 $\mu\text{g}/\text{m}^3$ (annual)	verified	Ministry of Environment, Korea
United Kingdom	20 $\mu\text{g}/\text{m}^3$ (annual, limit value in force since 1 Jan 2020; England target 10 by 2040)	verified	Air Quality Standards Regulations 2010, Sch. 2 (Stage 2 limit value); DEFRA National air quality objectives table, 2023-04-03
United States	9.0 $\mu\text{g}/\text{m}^3$ (annual, primary NAAQS, 2024)	verified	EPA Final Reconsideration of PM NAAQS, Federal Register 2024-02637

S6 Composite outcome and scoring rubric

Every response carries five subcomponents each normalised to $[0, 1]$. The factual subcomponent is decided by code where the registry holds a value (T1, and the resolvable part of T2 and T3); the three-vendor panel scores the residual of T2 and T3 and all of T4; T5 keeps the scores of the collection judge (GPT-5-mini for the pre-specified fifteen countries, Claude Haiku 4.5 for the ten added in the extension). The subcomponents are:

1. Factual accuracy against the registry value (binary for T1; partial credit for T2–T4).
2. Contextual completeness scored against a pre-validated rubric (T3, T4).
3. Citation quality, verified via DOI/official-source check.
4. Calibration, the alignment between stated confidence and correctness.
5. Absence of hallucination, coded binarily (0 hallucinated, 1 faithful).

The primary composite uses a pre-specified weighting (factual 0.30, contextual 0.25, citation 0.15, calibration 0.15, hallucination 0.15). As a sensitivity analysis the headline effects are recomputed under two alternative weightings — equal weights and a PCA-derived set (the first principal component of the five subcomponents) — and a conclusion is treated as robust only if its direction holds under all three; every headline effect keeps its sign under every weighting (Table S12). The composite is computed per (country, model, persona) cell to support the H6 difference-in-differences test. Each scored record also stores the judge’s free-text rationale; for example, the `gpt-5-mini` judge scored a Llama 4 Scout reply on Brazil’s T1 standard 0.15 with the rationale that it “gives wrong regulation (491/2018 vs correct 506/2024) and wrong current value... included a specific but incorrect citation and fabricated timing,” illustrating how the factual-accuracy and hallucination subcomponents discriminate substantive failure.

S7 Judge-panel reliability

Reliability is computed on every item the panel scored — the same items the reported effects are computed from — rather than on a stratified calibration subset. All 3,190 responses requiring a judge were independently scored by all three panel members, chosen for vendor diversity so that intra-vendor correlation does not inflate apparent agreement. The panel replaces the single-judge design of the original protocol, in which the primary judge (`gpt-5-mini`) was itself among the evaluated models.

Pairwise agreement is strongest between Claude and DeepSeek and lower for the systematically more lenient Gemini. Because every reported effect is a between-group contrast (tier gap, language penalty, task floor), a judge’s leniency offset cancels and does not threaten those contrasts — which is precisely why the analysis uses the panel *mean* rather than any single judge, whose reliability alone would be $\text{ICC}(2, 1) = 0.56$.

Table S8. Inter-judge reliability on all 3,190 panel-scored items. The operative quantity is the reliability of the panel *mean*, ICC(2,3); by the Spearman–Brown relation it is high even when single judges agree only moderately. Source: `panel_reliability_full.json`.

Quantity	Value
Judges (k)	3 (Google, Anthropic, DeepSeek)
Items scored by all judges	3,190
Krippendorff’s α (interval)	0.527
Single-judge ICC(2,1)	0.558
Panel-mean ICC(2,3)	0.791
Pairwise Pearson: Claude Sonnet DeepSeek-V3	0.801
Gemini 2.5 Pro Claude Sonnet	0.663
Gemini 2.5 Pro DeepSeek-V3	0.654
Mean composite by judge: Gemini 2.5 Pro	0.617
Claude Sonnet 4.6	0.451
DeepSeek-V3	0.379

S8 Statistical methods and full confirmatory test outputs

The three primary confirmatory tests form a single family (F1) corrected together with the Bonferroni–Holm step-down procedure ($\alpha = 0.05$). Secondary confirmatory tests are reported unadjusted within family; declared exploratory tests use Benjamini–Hochberg FDR at $q = 0.10$. Rank correlations use Spearman ρ ; monotonic-trend robustness uses the Mann–Kendall test; the persona effect uses a country-level permutation test (5,000 permutations). All outputs below are regenerated by `formal_tests.py`, `robust_tests.py` and `h4_corpus_mechanism.py` under `code/analysis/`.

Table S9. Primary family (Bonferroni–Holm): the pre-specified 15-country sample beside the 25-country extension the manuscript reports. Both columns are computed on the corrected scoring, from the same per-country accuracy that feeds the main text (`code/analysis/pre15_corrigido.py`). No primary test rejects its null in either sample, and the gradient stays below the pre-specified $\rho \geq 0.55$ threshold in both — so the locked protocol and the extension agree on the verdict, and the extension buys power rather than a different conclusion.

Test	15 countries (pre-specified)	25 countries (extension)
H1 Spearman $\rho(\text{acc}, \text{HDI})$	+0.079 ($p = 0.78$), ns	+0.361 ($p = 0.076$), ns
H1 Mann–Kendall (by HDI)	$S = 3$, $Z = +0.10$, $p = 0.92$, ns	$p = 0.065$, ns
H1 Spearman $\rho(\text{acc}, \text{Joshi})$	+0.131 ($p = 0.64$), ns	−0.061, ns (development, not linguistic)
H1 vs threshold $\rho \geq 0.55$	not met	not met
H4 partial $\rho(\text{acc}, \text{coverage} \text{HDI})$	+0.381 ($p = 0.18$), ns	+0.190 ($p = 0.37$), ns
H6 DiD persona (GS–GN)	not recomputed at $n = 15$	+0.48 pp, perm $p = 0.287$, ns
Tier gap (GN–GS), composite	+6.1 pp, $p = 0.12$ (3 GN)	+5.0 pp, $p = 0.019$

S9 Additional robustness and corroboration analyses

The headline effects were re-tested under several pre-specified analyses. All are regenerated by four scripts under `code/analysis/` (`weighting_and_evaluate`, `glmm_and_manipcheck`, `bayesian_reestimation`, `mediation_h4`), wired into `run_all.py` with the `--confirmatory` flag.

Table S10. H4 corpus mechanism, multi-measure (25 countries). Country-coverage breadth (Wikidata sitelinks) is the strongest of the three coverage proxies at the zero-order level ($p = 0.070$) and attenuates further after adjusting for development (HDI); we therefore report it as suggestive/exploratory, not as an established mechanism. The pre-specified language-edition proxy is null.

Measure	Family	$\rho(\text{acc, log measure})$	partial HDI
English-Wikipedia bytes	country	+0.146 ($p = 0.49$)	+0.125 ($p = 0.56$)
Wikidata sitelinks	country	+0.369 ($p = 0.070$)	+0.190 ($p = 0.37$)
Wikidata statements	country	+0.248 ($p = 0.23$)	+0.125 ($p = 0.56$)
Wikipedia-edition count	language	+0.048, ns	ns

Table S11. Robust secondary findings (25 countries), with leave-one-country-out (LOCO) and range-restriction checks.

Finding	Statistic
Tier gap (GN–GS)	+5.0 pp, 95% CI [+1.6, +8.4]; permutation $p = 0.019$
Tier gap, LOCO range	[+4.4, +6.0] pp, stable
Tier gap, extension wave only	+5.9 pp [+2.3, +9.6] (7 GN vs 3 GS), perm $p = 0.062$
Tier gap, code-adjudicated outcomes only (T1; resolved T2, T3)	GN 0.61 vs GS 0.46, +14.7 pp [+5.6, +23.9], perm $p = 0.012$; pooled OR 0.53 [0.48, 0.58] (T1 0.32, T2 0.51, T3 0.65)
H1 leave-one-country-out range	$\rho \in [+0.3, +0.5]$: never reaches the pre-specified 0.55
Task floor (T1+T2 vs T3–T5)	0.434 vs 0.612, Cliff’s $\delta = -0.41$
H3 Cabra-Mistral vs rest	0.351 vs 0.557, Cliff’s $\delta = -0.48$
H3 pre-specified (Cabra vs Llama 3.1 8B, Portuguese prompts)	Brazil 0.258 vs 0.313, $\delta = -0.26$ ($n = 20$ each); BRA+PRT+AGO 0.304 vs 0.362, $\delta = -0.17$ ($n = 60$ each)
H2 native vs English (all)	$\Delta = -5.0$ pp, Wilcoxon $p = 5 \times 10^{-16}$ ($n = 839$); mixed model, random intercept per country -5.0 pp ($p = 6 \times 10^{-7}$), per model -4.8 pp ($p = 2 \times 10^{-9}$); country as unit 8/9 negative, $t = -4.6$, $p = 0.002$; model as unit 12/14, Wilcoxon $p = 0.058$
Spanish / Portuguese / Hindi	-4.5 / -4.2 / -11.4 pp, all significant
by scoring instrument	code -3.8 ($n = 333$), panel -8.6 ($n = 314$), collection judge -1.1 pp ns ($n = 192$, T5)
code verdict only, by language	Portuguese -9.1 ($n = 132$, $p = 0.019$), Spanish -2.1 ($n = 165$, ns), all -5.0 pp ($n = 318$, $p = 0.048$)
H5 open vs closed-accessible	+12.8 pp closed advantage (descriptive)

Composite-weighting sensitivity. Recomputing every effect under the primary author-specified weighting, equal weights, and a PCA-derived weighting (first principal component of the five sub-components) leaves all four headline conclusions stable in direction (Table S12).

Mixed model and Bayesian re-estimation. A Gaussian linear mixed model (`statsmodels` MixedLM, REML) with crossed random intercepts for country and model gives a Global South effect of $\beta = -0.063$ (SE = 0.036, $p = 0.086$), marginal rather than conventionally significant because it charges the large between-country heterogeneity to the error term. A Bayesian hierarchical re-estimation with weakly informative priors (`pymc`; crossed country/model intercepts, `random_seed=42`) agrees on magnitude and is decisive about the sign: posterior mean -0.049 , 94% HDI $[-0.088, -0.006]$, $P(\beta < 0) = 0.987$ ($\hat{R} = 1.001$).

Table S12. Headline effects under three composite weightings (English confirmatory, $n = 6,573$). Direction is stable throughout; the HDI gradient stays below the pre-specified 0.55 under all three.

Weighting	Tier gap (pp)	$\rho(\text{HDI})$	Floor δ	H3 δ
Author-specified (primary)	+5.05	+0.361	−0.410	−0.476
Equal	+4.22	+0.355	−0.383	−0.507
PCA-derived	+4.72	+0.380	−0.435	−0.435

E-values. The tier gap carries an E-value of 1.71 at the point estimate and 1.31 at the confidence limit nearest the null (VanderWeele & Ding 2017 approximation from the standardized effect); the second is the operative number, and it is a low bar. No E-value is reported for the development gradient: its interval already includes the null, so the E-value at that limit is 1.00 and no confounder at all is required.

Exploratory mediation (H4). A standardized country-level path model (`semopy`, $n = 25$): $\text{HDI} \rightarrow \text{sitelinks}$ $a = +0.67$; $\text{sitelinks} \rightarrow \text{accuracy}$ $b = +0.27$; direct $c' = +0.31$; indirect $ab = +0.18$ with bootstrap 95% CI $[-0.27, +0.59]$ (includes zero). Country coverage is thus a suggestive but not established mediator, consistent with the main-text framing.

Persona manipulation check. Across 4,349 persona-condition responses, 50.2% explicitly acknowledge the public-manager role frame, so the null persona effect (H6) is not an artifact of the frame being universally ignored.

S9.1 Tier gap under alternative country partitions

The Global North/South split follows the UNCTAD classification. A gap visible under one taxonomy alone would be an artefact of that taxonomy, so we re-estimated it under three development-based partitions of the same 25 countries (Table S13). The gap is significant under UNCTAD and non-significant under every partition drawn on HDI, which is consistent with the null HDI gradient reported in the main text.

Table S13. Tier gap under alternative partitions of the same 25 countries. Permutation test with country as the unit, 10,000 relabellings.

Partition	Gap (pp)	GN/GS	p
UNCTAD developed/developing (used throughout)	+5.05	10/15	0.019
HDI below the sample median (0.781)	+2.14	13/12	0.329
HDI < 0.800 (UNDP “very high” cutoff)	+2.78	12/13	0.206
HDI < 0.700	+1.94	20/5	0.482

S10 Deviations from the pre-specified analysis plan

The analysis plan required that departures be logged and classified, and set a ceiling on them: confirmatory status is retained only if deviations affect no more than 15% of planned cells. The table below reports every departure we identified. The delivered scope is 27.5% of the

planned response count, so the ceiling is exceeded by a wide margin, and the study is reported as exploratory throughout, as described in the Study protocol statement.

Table S14. Departures from the pre-specified plan, with the effect of each on interpretation. “Planned” quotes the analysis plan; “Delivered” is what the released artefacts contain.

Item	Planned	Delivered	Effect on interpretation
Response count	33,600 responses (600 prompts $\times$ 14 models $\times$ 2 personas $\times$ 2 replicates)	9,251 judge scores over 8,300 cells (27.5% of the planned count), 8,244 analysed	Triggers the plan’s own 15% ceiling. The study is reported as exploratory.
Public registration	Plan deposited in a public registry before confirmatory collection	Plan fixed in a version-controlled document; never deposited	Timing cannot be verified independently of repository history. We do not claim pre-registration.
Rubric pilot	50-prompt pilot scored by two independent human raters, Krippendorff’s $\alpha \geq 0.70$ required before scaling	Not run. No rater-level α exists	Rubric reliability is evidenced only after the fact, on confirmatory data, at $\alpha = 0.527$ across the 3,190 items scored by all three panel judges, below the threshold the plan had set.
Judging	Three-judge LLM ensemble plus external human annotators blind to hypotheses	Three-vendor panel on every item that required a judge (3,190 items); code adjudication where the answer is a checkable register value; no human layer	The ensemble is as planned, but no human gold standard exists anywhere in the study. Reliability rests on model-to-model agreement, $ICC(2, 3) = 0.79$.
Ground-truth registry	Per (country, task) entry with source class, resolvable URL, gold value, reference date, human validator and validation flag	Registry covers T1 for all 25 countries; T2, T3 and T4 were built in a later revision from single international sources (WHO AAQD v6.1, WHO GHO AIR_41, UNEP GAAPL Appendix 1); no human validator field exists	Ground-truth provenance is now anchored for four of five tasks but is not uniform in source class, and the Methods now describe the four registries as built.
Country sample	15 countries, fixed in advance	15 extended to 25 after the pre-specified analysis was run	The extension was made to gain power for the gradient, which nonetheless does not reach significance at $n = 25$ under the corrected scoring ($\rho = 0.36$, $p = 0.076$). The primary family and the tier gap are reported alongside their 15-country values (Table S9).

Item	Planned	Delivered	Effect on interpretation
Scoring instrument	Single primary judge (GPT-5-mini) scoring all tasks against the registry	The collection judge was not single: GPT-5-mini scored the pre-specified fifteen countries and Claude Haiku 4.5 the ten extension countries, which hold seven of the ten Global North countries, so judge and tier were confounded in the judged scores. T1, T2 and T3 are now adjudicated by code against official registers where the returned value is extractable (99.8%, 56.3% and 71.8% of cells); the residual and all of T4 re-scored by the three-vendor panel mean; T5 keeps the collection judges' scores	Changes the substantive conclusions on the same responses. The development gradient falls from $\rho = 0.51$ ($p = 0.004$) to $\rho = 0.36$ ($p = 0.076$); the native-language penalty grows from -2.1 pp to -5.0 pp and becomes significant in all three languages; the T1 odds ratio moves from 0.22 under the judges to 0.32 under code. The findings are reported in the revised order.
T1 key	Value in force per national instrument	United Kingdom key corrected on 2026-09-21 from 10 (English target for 2040) to $20 \mu\text{g}/\text{m}^3$ (limit value in force since 2020); Egypt has no retrievable value and is excluded from T1; Bangladesh corrected from the 2005 gazette (15) to the 2022 Rules (35) after external review; official ladders (Brazil, EU, UK, US, Canada, Australia, Colombia, Bangladesh) stored alongside the value in force; registry versions carry a SHA-256 prefix	Tier comparison on T1 reported with and without the three no-standard countries (0.32 and 0.39), and under the ladder-tolerant key (0.14).
Ordering of findings	Plan treated H1 (geographic gradient) as the primary hypothesis	H2 (native-language penalty) is reported as the principal finding and H1's gradient as exploratory	Post-hoc reordering, driven by the corrected scoring rather than by the data being re-examined for a better story. Both hypotheses were pre-specified and both are reported; only the prominence changed.

Item	Planned	Delivered	Effect on interpretation
Judge prompt defect	—	An early T2 judge prompt asked the judge to compare a value against a tolerance band expressed in words, requiring it to do the arithmetic. Concordance on those items was $r = -0.04$; pre-computing the range raised it to $r = +1.00$	The 274 scores collected under the ambiguous wording were discarded and recollected. Reported because the defect first presented as genuine inter-judge disagreement.
Corroborating analyses	Mixed model, Bayesian re-estimation and E-values reported alongside the primary tests	Recomputed on the corrected scoring	All three are weaker than the single-judge versions. The mixed model falls from $p = 0.007$ to $p = 0.086$; the Bayesian posterior from $P(\beta < 0) = 1.00$ to 0.987 ; the tier-gap E-value from 1.85 to 1.71 at the point and 1.31 at the confidence limit. The tier gap is reported as supported but specification-sensitive.
E-value reporting	E-value of the point estimate for tier gap and HDI gradient	E-value of the point estimate <i>and</i> of the confidence limit nearest the null, for the tier gap only	Corrects a defect of our own. Reporting only the point estimate overstates robustness. The gradient's E-value is withdrawn: the association is not significant, so its nearest-null E-value is 1.00 and no confounder is required to explain it away.
H4 adjustment set	Partial Spearman adjusting for HDI <i>and</i> log GDP per capita	Originally adjusted for HDI only; GDP did not appear in the analysis code. Now executed as specified	Resolved. The specified adjustment gives $\rho = +0.02$ against $+0.04$ with HDI only; the two control sets agree because HDI and log GDP correlate at $\rho = 0.97$ across these countries.

Item	Planned	Delivered	Effect on interpretation
Native-language rendering	Human native-speaker translation with independent back-translation and bilingual adjudication	LLM translation, audited after the fact by a back-translation census of all 90 native prompts: a different model back-translates, two further models audit against the English original on five items, one flag marks the prompt divergent	Partially resolved. No human translator and no bilingual adjudication, so the plan is not met. But the artefact the plan guarded against is measured and absent: 90/90 preserve the country and the requested quantity, 88/90 the question, and H2 is unchanged among verified-faithful prompts (−4.8 pp; fidelity does not predict effect size, permutation $p = 0.80$).
Extra H2 variants	Two additional question variants per (country, task), rendered in both languages	Generated but never collected or judged	H2 rests on the principal pairs only; the planned cell count for H2 was not reached.
Model coverage	All 14 stacks across all countries and tasks	Coverage is not uniform: two stacks are absent from several countries after provider rate limiting and credit exhaustion	Per-country means are computed over different model sets; the leave-one-model-out re-estimates of H2 (Section 4 of the main text) and the per-model N in Table S1 are the checks reported.

A further property of the response corpus, not a departure from the plan but a consequence of how collection was retried, belongs here because it bears on reproducibility. Of 9,760 unique (model, prompt, replicate) keys, 2,147 (22.0%) have more than one stored response, produced by re-runs on different dates; restricting to records the scoring pipeline admits gives 9,140 keys and 2,095 duplicates (22.9%). Earlier drafts reported 4,976 and 1,003 here, figures that pre-date the extension to 25 countries and that we could not reproduce from the released artefacts; the counts above are regenerated by `code/analysis/robustness_extra.py`. The stored answers differ substantially between runs; in one case the same prompt and model yielded 307 and 3,180 characters. The answer that enters the analysis is the first stored in run-file order, which is the one the collection judge scored (the judge runner reads the run files in order and skips keys already scored); the code verdicts for T1, T2 and T3 are computed on that same answer, so both instruments describe one response. Scoring every stored answer and averaging the verdicts instead changes the code verdict in 3.1% of cells and the mean factual score by 0.0002(`code/analysis/t1_and_wave_sensitivity.py`). Re-judging under a chronological rule would require re-running the panel and was not done.

Extractor audit for T1. All ten native-language T1 responses that the extractor classified as carrying no comparable value were read one by one (`data/processed/t1_extractor_audit.md`). None contains an annual PM_{2.5} value in $\mu\text{g}/\text{m}^3$ that the extractor missed: seven are degenerate

outputs of the regional 7B model (repeated prompt, nonsense text), two cite the regulation without a number, and one asks permission to look the value up; two further responses state “15 mg/m³”, a value in the wrong unit, which is treated as no comparable value in every language. The extractor’s handling of decimal commas, Devanagari numerals and units written in words is fixed by 13 unit tests released with the code. The audit was performed by the analysis pipeline’s author rather than by an independent reader, and is recorded as such.

S11 Prompt construction, persona template, and verbatim examples

S11.1 Representative failure modes, verbatim

Table S15. Box 1 — Representative failure modes (verbatim, lightly truncated).

Failure mode	Example
Fabricating a standard that does not exist	Angola has no national norm. Gemini 2.5 Flash asserted an annual standard of “25 $\mu\text{g}/\text{m}^3$ ” in one replicate and “15 $\mu\text{g}/\text{m}^3$ ” in another. GPT-5 asserted “15 $\mu\text{g}/\text{m}^3$ ” for Nigeria, which regulates PM ₁₀ only (NESREA). A faithful response would decline to state a value that does not exist.
Missing the binding value (the T1 floor)	Brazil’s official annual standard is 17 $\mu\text{g}/\text{m}^3$ (CONAMA 506/2024). The modal answer was 15 $\mu\text{g}/\text{m}^3$ (42 responses), with 10, 20 and 25 also common; the official value was almost never returned.
Degradation in the native language (H2)	India’s annual standard is 40 $\mu\text{g}/\text{m}^3$. GPT-OSS 120B returned the correct value in English and an <i>empty</i> answer to the same question in Hindi.

Prompts were authored in English for the air pollution policy domain with domain supervision (PPGSAU/UTFPR) and validated against the ground-truth registry before entering the confirmatory stimulus set. The plan required a two-rater pilot to reach Krippendorff’s $\alpha \geq 0.70$ before scaling; that pilot was not run, and the deviation is logged in Section S10. Native-language renderings (Portuguese, Spanish, Hindi) were produced by a high-quality LLM translator (Claude Sonnet 4.6) instructed to preserve the meaning, the persona frame, and proper nouns/acronyms, with the factual ground truth held in English (language-invariant). Human native-speaker translation was not performed, and this is noted as a limitation in the main text; an LLM back-translation census of all 90 native prompts, by a model other than the translator, is reported in Section S10 and in the main text (Section 4). Every factual claim in a ground-truth entry was required to resolve to an official-source URL or a peer-reviewed DOI before the corresponding prompt was admitted (the project Citation Verification Protocol applied at the ground-truth generation stage, not only at the manuscript stage).

Persona manipulation. Persona is a within-prompt factor with two levels that hold the substantive question identical. The **neutral** level presents a plain information request; the **public_manager_env** level prepends a fixed role frame. The contrast is shown verbatim for Brazil, T1:

Neutral. “What is the current national annual mean ambient air quality standard for PM2.5

in Brazil? State the regulation that establishes it and whether it is an intermediate or final target. Limit to 1–3 sentences.”

Public environmental manager. “I am the municipal environmental management official of São Paulo, Brazil, and I need to brief decision-makers. What is the current national annual mean ambient air quality standard for PM_{2.5} in Brazil? State the regulation that establishes it and whether it is an intermediate or final target. Limit to 1–3 sentences.”

The corresponding registry ground truth is: “Brazil’s national PM_{2.5} standards are set by CONAMA Resolution 506/2024 (which superseded the relevant articles of Resolution 491/2018). The stage in force in 2026 is Intermediate Target PI-2; the Final Target adopts the WHO 2021 annual guideline of 5 $\mu\text{g}/\text{m}^3$ ” (source: CONAMA Resolution 506/2024). Native-language prompts apply the same template, translated; for example the Hindi T1 prompt for India is the faithful Devanagari rendering of the neutral English item, reproduced verbatim in the released prompt set. The complete prompt set (370 items: 250 English prompts, 25 countries $\times$ 5 tasks $\times$ 2 personas; 90 native-language renderings; and 30 additional question variants generated at design time and never collected) is released as `prompts_confirmatory.jsonl` with the per-item ground truth, source, and rubric.

References

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